

Inventory Management System

Project Requirements & Database Design Specification

1. Project Overview

The Inventory Management System is a web application designed to help a business manage its products, suppliers, purchases, sales, and stock. The primary objective of the system is to provide an easy, reliable, and automated way to track inventory and keep product quantities up to date in real time.

2. Main Features

Product Management

- **Add Products:** Add new products to the inventory database.
- **Edit Products:** Edit existing product information, pricing, and details.
- **Delete Products:** Delete products from the system.
- **View Details:** View complete details of a specific product.
- **Browse Stock:** Browse all available products in dynamic lists.
- **Pagination:** Display a suitable number of products in each page.
- **Search & Filter:** Search and filter products by name, category, or status.

Category Management

- **Manage Categories:** Add, edit, and delete product categories.
- **Assign Products:** Assign products to appropriate categories.
- **Category View:** View products contained within a specific category.

Supplier Management

- **Manage Suppliers:** Add and manage supplier profiles.
- **Contact Information:** Store contact information for each supplier (phone, email, address).
- **Supplier Products:** View and map the list of products supplied by each vendor (Many-to-Many relationship).

Inventory Management

- **Stock Tracking:** Track the current quantity of each product in real time.
- **Purchase Adjustments:** Automatically increase stock when new products are purchased.
- **Sales Adjustments:** Automatically decrease stock when products are sold.
- **Low-Stock Alerts:** Display products with low stock levels for timely replenishment.

Purchase Management

- **Create Purchases:** Create new purchase records for stock replenishment.
- **Supplier Selection:** Select authorized suppliers for order processing.
- **Add Line Items:** Add products and specify purchase quantities.
- **Total Calculation:** Calculate the total purchase amount automatically.
- **Purchase Details:** View detailed purchase transaction history.

Sales Management

- **Create Sales:** Create sales records for customer orders.
- **Add Sold Products:** Add products and quantities to individual sales transactions.
- **Total Calculation:** Calculate the total sale amount automatically.
- **Automated Stock Deduction:** Automatically update product stock upon completing a sale.
- **Sales Details:** View detailed sales receipts and transaction history.

3. Dashboard & Statistics

The system includes a centralized Dashboard page designed to display actionable visual statistics and summaries. This provides business owners with instant clarity regarding overall inventory health, sales velocity, and operational performance.

Key System Metrics & Analytics

- **Total Products:** Total count of cataloged items in the database.
- **Total Categories:** Total number of active product categories.
- **Total Suppliers:** Total count of registered suppliers.
- **Total Stock Quantity:** Aggregated physical inventory items stored across all products.
- **Low-Stock Products:** Alert count highlighting items below their defined reorder threshold.
- **Total Purchases:** Cumulative count and total monetary value of inventory purchases.
- **Total Sales:** Cumulative count and overall revenue generated from sales.
- **Recent Activity Log:** Real-time feed showing recent sales and incoming stock purchases.
- **Most Sold Products:** Top-performing items ranked by total quantity sold and revenue generated.

DESIGN RECOMMENDATION Note: Students should choose suitable visual components (such as KPI cards, summary tables, and analytical charts) that effectively help business owners assess inventory performance and make informed replenishment decisions.

4. Database Architecture & Design

Below is the updated relational database schema specification including the SupplierProducts junction table to support the many-to-many relationship between Suppliers and Products.

Entity Name	Key Attributes	Description & Relationships
Categories	CategoryID (PK), CategoryName, Description	Stores product categories. Has a 1-to-Many relationship with Products.
Suppliers	SupplierID (PK), SupplierName, ContactName, Phone, Email, Address	Stores vendor contact records. Has a 1-to-Many relationship with Purchases and SupplierProducts.
Products	ProductID (PK), SKU, ProductName, CategoryID (FK), UnitPrice, StockQuantity, LowStockThreshold	Core product inventory entity. Linked to Categories, Purchase Items, Sales Items, and SupplierProducts.
SupplierProducts	SupplierProductID (PK), SupplierID (FK), ProductID (FK), SupplierSKU, ContractPrice, LeadTimeDays	Junction table representing the Many-to-Many relationship between Suppliers and Products. Allows multiple suppliers for a product and vice versa.
Purchases	PurchaseID (PK), SupplierID (FK), PurchaseDate, TotalAmount	Header table tracking overall procurement orders from suppliers.
Purchase Items	PurchaseItemID (PK), PurchaseID (FK), ProductID (FK), Quantity, UnitCost	Junction table storing itemized details for each purchase order.
Sales	SaleID (PK), SaleDate, TotalAmount, CustomerInfo	Header table recording customer sales transactions.
Sales Items	SaleItemID (PK), SaleID (FK), ProductID (FK), Quantity, UnitPrice	Junction table linking sold products with their specific sales order.

5. Generative AI Integration

The system must include at least one meaningful Generative AI feature that adds value to the Inventory Management System. The AI feature should solve a real problem or improve the user experience.

- **AI Chatbot / Inventory Assistant:** Allow users to ask questions about products, stock, suppliers, purchases, or sales using natural language.
- **RAG (Retrieval-Augmented Generation):** Enable the AI to retrieve relevant information from system data or uploaded business documents and provide grounded answers.
- **Natural Language Analytics:** Allow users to ask questions such as 'What are the most sold products this month?' or 'Which products are currently low in stock?'
- **AI-Powered Recommendations:** Suggest products that should be restocked based on inventory levels, sales history, or other available data.
- **AI Report Generation:** Generate summaries or insights from sales, purchases, and inventory statistics.
- **Document / Knowledge Assistant:** Allow the AI to answer questions based on uploaded inventory-related documents, policies, product manuals, or supplier information.